



AERSENTINEL OPERATIONAL MANUAL

Standard Operating Procedures & Complete User Manual for Meteorological Quality Control, Fleet Health Monitoring, and Predictive Sensor Maintenance

Document ID:	AERO-SOP-2026-v2.4
System Version:	AeroSentinel v2.4-PROD (Enterprise Release)
Problem Statement:	Smart India Hackathon (SIH 26073)
Target Audience:	IMD Duty Forecasters, QC Analysts, Field Technicians, Quality Officers, Station Operators
Classification:	Official Meteorological Operations Guide (IMD / WMO)
Published:	September 2026

INFO: This manual outlines the operational workflows for the AeroSentinel Automatic Weather Station (AWS) Quality Control platform. Personnel must execute triage and adjudication following the protocols herein.

1. System Overview & Core Objectives

AeroSentinel is an intelligent meteorological monitoring and quality control system built to address SIH 26073. Automatic Weather Stations (AWS) deployed across regional networks frequently experience sensor degradation, calibration drift, communication dropouts, and environmental interference. AeroSentinel safeguards national weather observation networks through automated multi-tier anomaly detection, geospatial physical cross-validation, and closed-loop technician dispatch.

Key Operational Benefits:

- **Dual-Risk Mitigation:** Prevents corrupted sensor data from tainting Numerical Weather Prediction (NWP) models while avoiding false-alarms during genuine regional severe weather events.
- **Multi-Tier Decision Fusion:** Combines deterministic physical laws (Tier 1), ML Isolation Forests and LSTM-Autoencoders (Tier 2), and 3D KDTree Geospatial Neighbor Validation (Tier 3).
- **Explainable Verdicts:** Every flagged reading includes transparent reason codes, confidence percentages, and SHAP feature attribution metrics.
- **Self-Healing Imputation:** Automatically substitutes corrupted sensor readings with Inverse Distance Weighted (IDW) spatial estimates or autoregressive temporal baselines.
- **Predictive Maintenance:** Ranks stations by 30-day failure probability and prescribes proactive field maintenance tickets before station downtime occurs.

2. Getting Started & User Authentication

AeroSentinel implements enterprise Role-Based Access Control (RBAC). All API requests and UI sessions are secured via cryptographic JSON Web Tokens (JWT) signed with HMAC-SHA256.

Pre-configured Demonstration Personas:

For immediate operational evaluation, the top header includes a quick **Role Switcher** dropdown. Operators can switch between the following 5 authenticated personas:

Role Name	Default User	Key Responsibilities & Permissions
Administrator	Admin User admin@aerosentinel.gov.in	Full root control: User management, RBAC configuration, system settings, global audit inspection.
Duty Forecaster	Dr. Rajesh Sharma forecaster@aerosentinel.gov.in	Fleet map oversight, live meteorological telemetry stream, synoptic weather analysis, extreme weather tracking.
QC Analyst	Pooja Nair qc@aerosentinel.gov.in	Anomaly adjudication queue, SHAP feature review, false-positive feedback labeling, ML active learning triggers.
Field Technician	Vikram Singh tech@aerosentinel.gov.in	Predictive maintenance queue, hardware health metrics, physical sensor work order dispatch, calibration logs.
Viewer	Public Observer viewer@aerosentinel.gov.in	Read-only access to fleet map, public dashboards, and station summary cards.

BEST PRACTICE: To switch between operational roles during demonstration, click on your profile pill in the top header and select the target role from the dropdown menu. The UI will instantly morph its navigation and operational tools.

3. Navigation Architecture & Layout Anatomy

AeroSentinel features a streamlined desktop & mobile layout designed for high-density 24/7 meteorological operations.

1. Top Application Header

- **Logo & Brand Badge:** Clicking the logo returns to your role's primary dashboard.
- **System Health Indicator:** Displays real-time operational status (OPERATIONAL, DEGRADED, CHECKING). Clicking navigates directly to System Health.
- **Active Alerts Bell:** Displays real-time badge count of unresolved critical/high alerts with pulse animation. Clicking opens the Alerts Feed.
- **Theme Toggle:** Switches instantly between Dark Meteorological Mode (default) and Light Mode.
- **User Profile & Role Pill:** Shows current authenticated user, avatar, and active role badge.

2. Collapsible Left Navigation Sidebar

The left sidebar organizes the platform into four logical operational zones. Items are strictly rendered based on the user's active RBAC permissions:

Category	Sidebar Item	Primary Function & Objective
Overview	Dashboard Fleet Map	Role-specific KPIs, triage queues, active fleet map, and synoptic network metrics.
Monitoring	Live Stream Stations Alerts Health	Real-time telemetry stream, station registry & time-series analysis, live alert triage feed, and service health monitors.
Operations	Tasks Upload Data Maintenance	Role-assigned operational workflows, batch CSV telemetry ingestion, and 30-day predictive failure risk console.
Administration	Users Roles & Perms Audit Log System Settings	User accounts, granular permission matrix, tamper-evident governance audit trail, and QC algorithm threshold tuning.

4. Step-by-Step Module Operational Guide

4.1 Fleet Map (Geospatial GIS Console)

The Fleet Map provides an interactive GIS view of all Automatic Weather Stations across the National Capital Region (NCR) and national grid.

- **Basemap Controls:** Select between CARTO Dark Matter (default), Positron (light), OpenStreetMap, or Offline Cached Tiles (`asset/map`).
- **Marker Status Legend:** Green = Normal (all sensors valid); Amber = Suspect (minor ML/spatial deviation); Red = Anomalous (confirmed fault); Gray = Stale (missing telemetry).
- **Station Detail Popup:** Clicking any station marker displays real-time temperature, humidity, pressure, wind speed, elevation, and health score, with a direct link to `Inspect Station Telemetry`.

4.2 Live Telemetry Stream

The Live Stream provides a real-time WebSocket view of incoming AWS telemetry packets arriving from weather sensors.

- **Stream Controls:** Use the **Pause / Resume** button to halt the incoming stream when analyzing a transient anomaly.
- **Variable Tabs:** Switch between Temperature (°C), Relative Humidity (%), Atmospheric Pressure (hPa), Wind Speed (m/s), and Precipitation (mm).
- **Status Indicators:** Each incoming packet shows its QC verdict badge (`VALID`, `SUSPECT`, `ANOMALOUS`) and timestamp.

4.3 Stations & Telemetry Inspection

Drilling into any station opens the full telemetry inspection dashboard:

- **Time-Series Chart:** Displays historical readings plotted against WMO upper and lower physical threshold boundaries.
- **Interactive Anomaly Tooltip:** Hover over any data point to inspect the exact QC verdict, reason code (e.g. `RULE\_FLATLINE`), fault type, and confidence score.
- **QC Diagnostics Table:** Below the chart, review flagged observations, reason codes, and automated self-healing imputed values.

4.4 Alerts Feed & Human-in-the-Loop Feedback

The Alerts Feed is the mission-critical interface where QC Analysts triage anomalies and train the AI model.

- **Severity Badges:** Critical (red), High (amber), Medium (yellow), Low (blue).
- **One-Click Acknowledgment:** Click **Acknowledge** to claim an alert and inform colleagues you are triaging the station.
- **Feedback Buttons:**
 - **Confirm Fault:** Labels the alert as a true sensor failure (e.g. flatline, broken heating element). Dispatches maintenance.
 - **Mark Genuine / False Alarm:** Informs the system that an unusual reading was actually caused by an extreme atmospheric event (e.g. squall).
- **Active Learning Retrain Console:** Located in the **Model Audit** sub-tab. Shows collected feedback samples. When ≥ 10 samples accumulate, click **Trigger ML Retraining** to retrain the Isolation Forest model on human feedback!

4.5 Predictive Maintenance & Field Dispatch

Predictive Maintenance calculates 30-day failure probabilities for every station across the network.

- **Risk Ranking:** Stations are ranked from highest to lowest risk (Critical $\geq 70\%$, Elevated 40–70%, Moderate 20–40%, Nominal $< 20\%$).
- **Primary Risk Driver:** The AI explicitly identifies **why** the station is predicted to fail (e.g., `Sensor Drift Accumulation`, `Sensor Flatline / Freeze`, `Hardware Lifespan Aging`).
- **Prescribed Action:** Shows actionable maintenance instructions (e.g. `Inspect solar charge controller`, `Recalibrate temperature transducer`).
- **Dispatch Ticket:** Click **Dispatch Ticket** to generate an official IMD Maintenance Work Order and deploy a field technician.

4.6 Batch Data Upload (CSV Ingestion)

Allows operators to upload historical or offline AWS data logs in standard CSV format:

- Drag-and-drop or browse for `.csv` files containing headers: `station\_code`, `timestamp`, `temperature`, `relative\_humidity`, `pressure\_hpa`, `wind\_speed`.
- AeroSentinel parses, validates, and runs all 4 QC tiers automatically in the background, updating station records immediately.

4.7 Administration & Immutable Audit Trail

- **User Management:** Admins can create new operator accounts, assign roles, and revoke access.
- **Roles & Permissions:** Interactive matrix showing all 15 permission nodes across the 5 system roles.
- **Audit Log:** Immutable record of every administrative action, alert feedback, model retrain event, and system configuration change, complete with user email, timestamp, and IP address.

5. Step-by-Step Operator Case Studies

Scenario A: Triaging a Temperature Sensor Spike Alert

1. Navigate to **Monitoring** → **Alerts** in the sidebar.
2. Locate the alert for station `NCR001` reporting an anomalous temperature reading of `58.2°C`.
3. Click **Inspect Station** to review the historical time-series chart and examine neighboring station temperatures on the Fleet Map.
4. If neighboring stations report normal temperatures (`34°C`), the reading is an isolated sensor spike. Click **Confirm Fault**.
5. The system confirms the fault, logs the feedback into the active learning pool, and marks the reading as anomalous in the QC database.

Scenario B: Preventing a False Alarm during a Regional Heatwave

1. A high-severity alert triggers for station `NCR005` with reason code `ML\_BORDERLINE` due to an unprecedented temperature of `47.5°C`.
2. Open the **Fleet Map** and examine the surrounding peer AWS stations (`NCR004`, `NCR006`).
3. All surrounding stations also indicate elevated temperatures (`46.8°C`, `47.1°C`), verifying a synoptic heatwave.
4. In the Alerts Feed, click **Mark Genuine / False Alarm**. Add notes: *'Regional heatwave confirmed by peer AWS'*.
5. The QC verdict is updated to `SPATIAL\_VALIDATED\_EXTREME`, ensuring genuine meteorological data is NOT lost from weather models!

Scenario C: Dispatching Field Maintenance for Calibration Drift

1. Navigate to **Operations** → **Maintenance**.
2. Observe the station ranked #1 in Critical Risk (e.g. `NCR007 - Rohtak`, $P(\text{failure}) = 84\%$).
3. Inspect the Primary Risk Driver: `Sensor Drift Accumulation`.
4. Click **Dispatch Ticket**. Review the technician checklist and click **Confirm & Dispatch Ticket**.
5. The work order is logged to the system with a unique ticket ID and dispatched to field technicians.

6. Frequently Asked Questions & Troubleshooting

Q: What happens if an AWS goes offline?

A: If an AWS fails to report data for 3 consecutive intervals (45 minutes), its status marker turns gray (`STALE`) on the Fleet Map, and its predictive maintenance risk score increases due to telemetry dropouts.

Q: Can I download data for external analysis in Python or R?

A: Yes. In the Station Detail view or Ingestion console, click **Export CSV** to download clean or raw telemetry with full QC verdict annotations.

Q: How does the Self-Healing Network correct bad readings?

A: When a sensor reading is flagged anomalous, AeroSentinel automatically calculates an imputed value using Inverse Distance Weighting (IDW) from the 5 nearest healthy peer stations. Downstream NWP models receive the corrected value along with an imputation confidence score.